Cloud Computing

Services "managés" /
Platform-as-a-Service

Plan / Services managés (PaaS)

> Catalogue de services

> Exemples de services managés

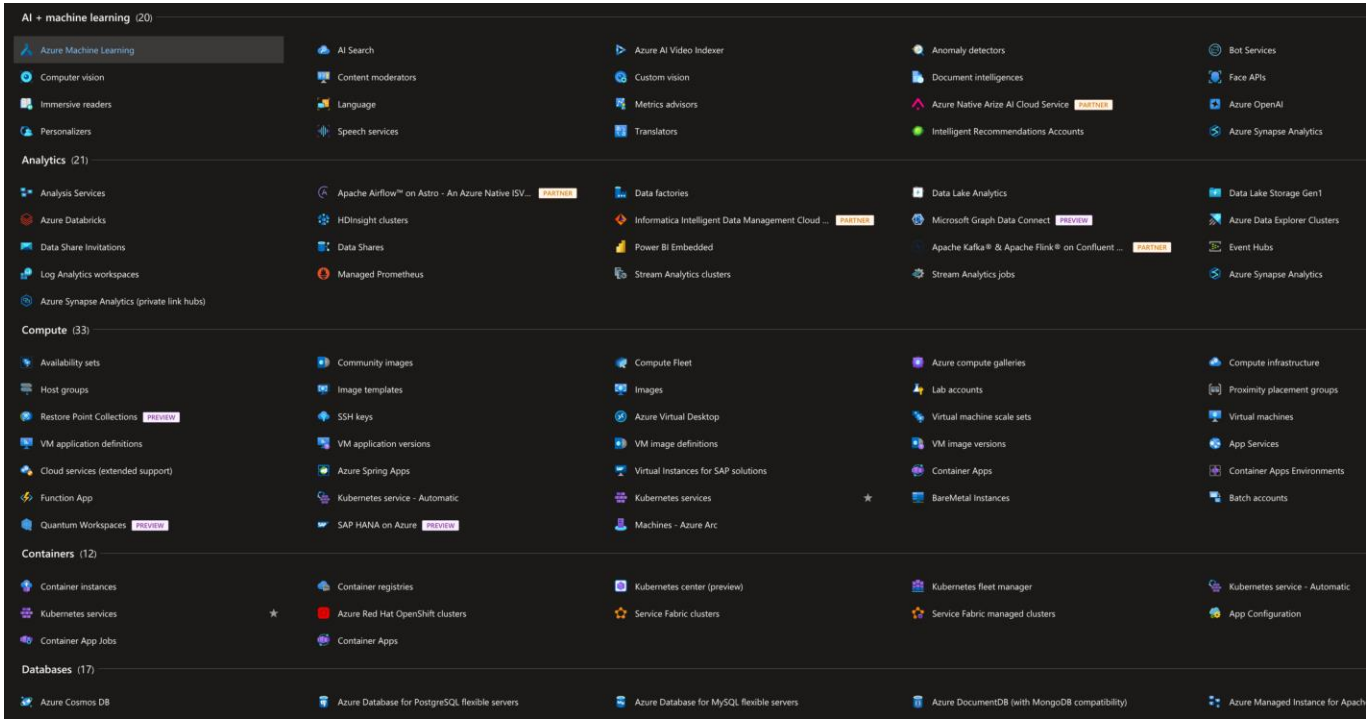
Azure – catalogue de services

350+ services

Familles de services

- Calcul, conteneurs, réseau, stockage
- AI, ML, Analytics
- Bases de données
- DevOps
- Hybridation, multicloud
- Identités, gouvernance
- Intégration
- IoT
- Observabilité
- Sécurité
- Services web (API, hébergement, …)

Azure – catalogue de services



https://portal.azure.com/#allservices/category/All

Plan / Service managés (PaaS)

> Catalogue de services

> Exemples de services managés

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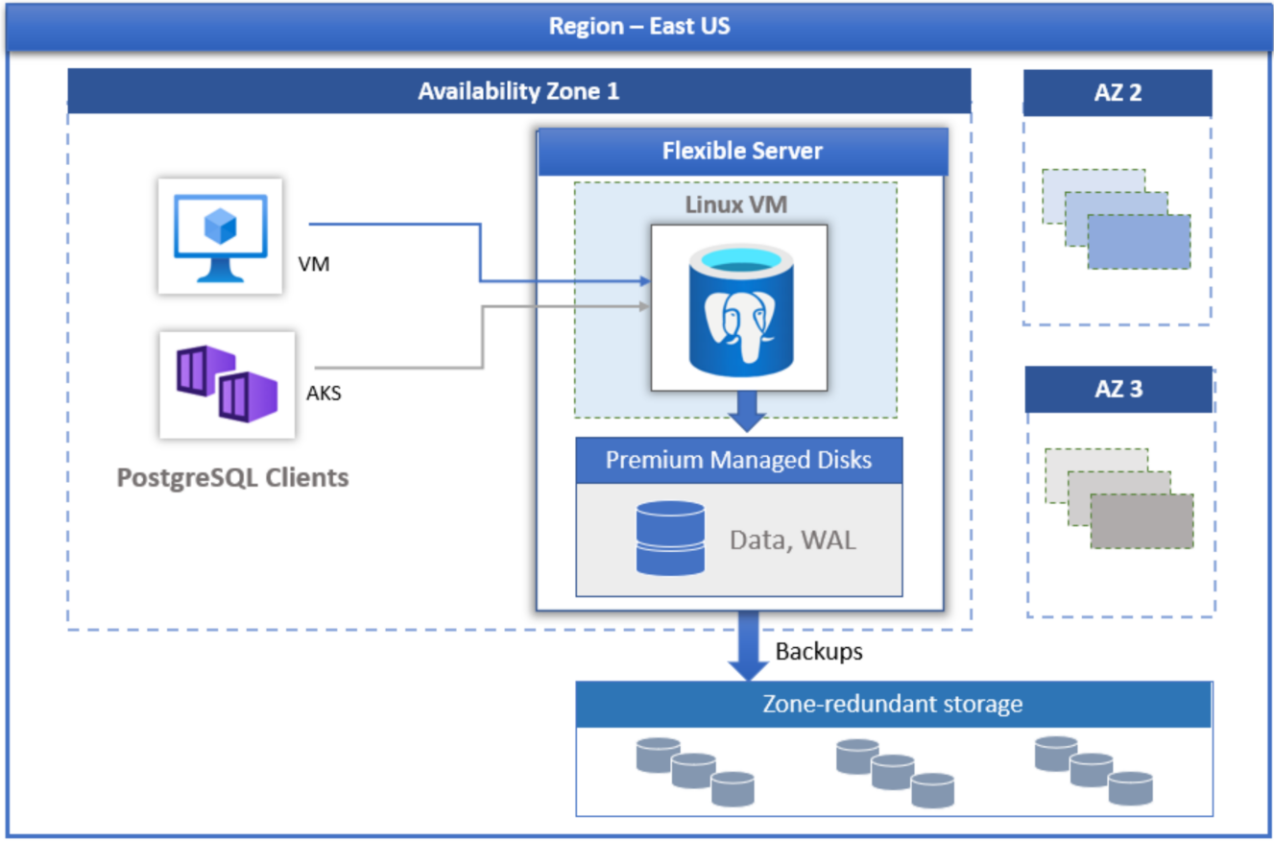
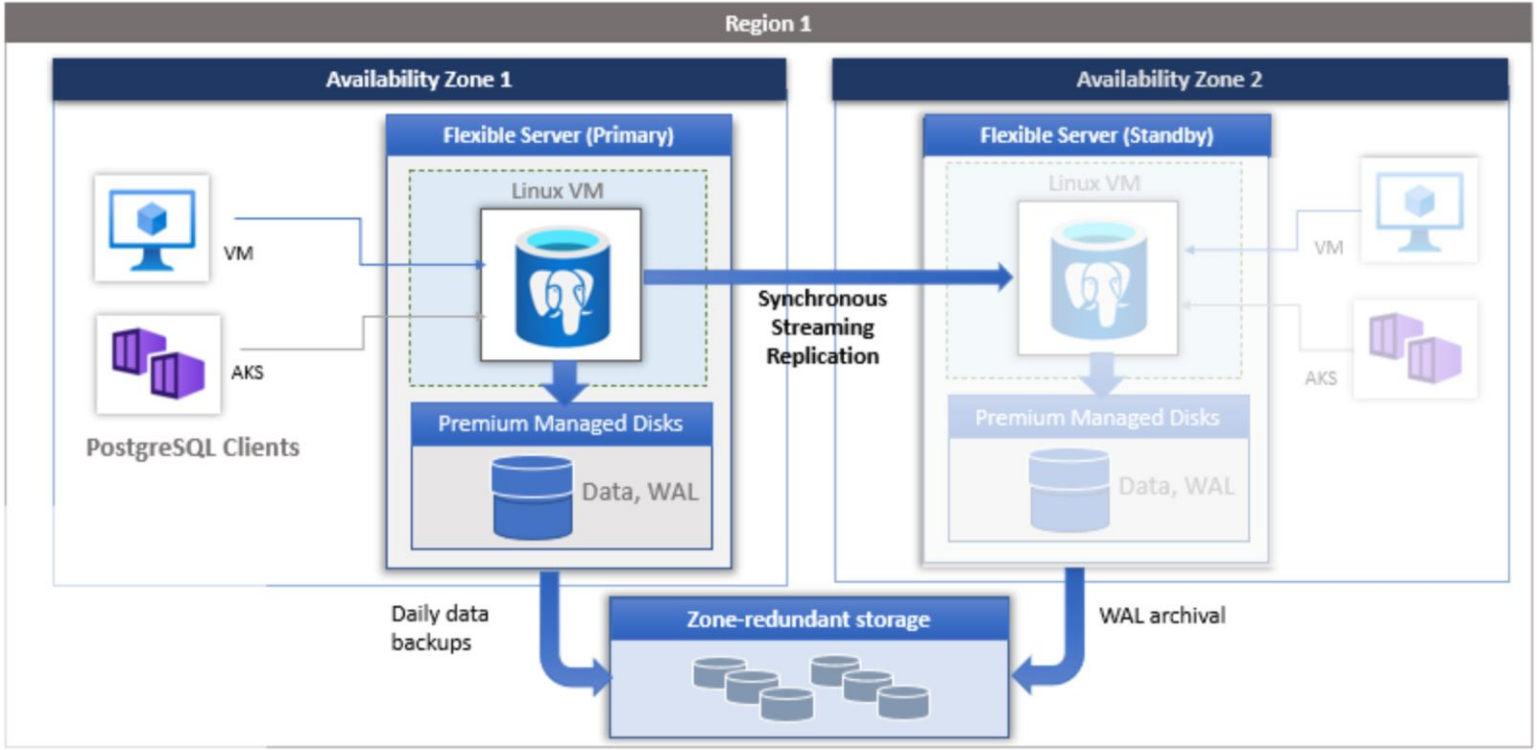
- **PostgreSQL**
- Kubernetes services
- Machine Learning

PostgreSQL Flexible Server (1/2)



PostgreSQL Flexible Server (2/2)



PostgreSQL Flexible Server – gestion facile

```
az postgres flexible-server create --resource-group myresourcegroup --name mydemoserver-pgsql --location eastus --admin-user myadmin --admin-password <password> --sku-name Standard_D4ds_v5 --tier GeneralPurpose --public-access 0.0.0.0 --storage-size 128 --tags "Environment=Development"
```

```
az postgres flexible-server upgrade --resource-group <resource_group> --name <server> --version <target_version>
```

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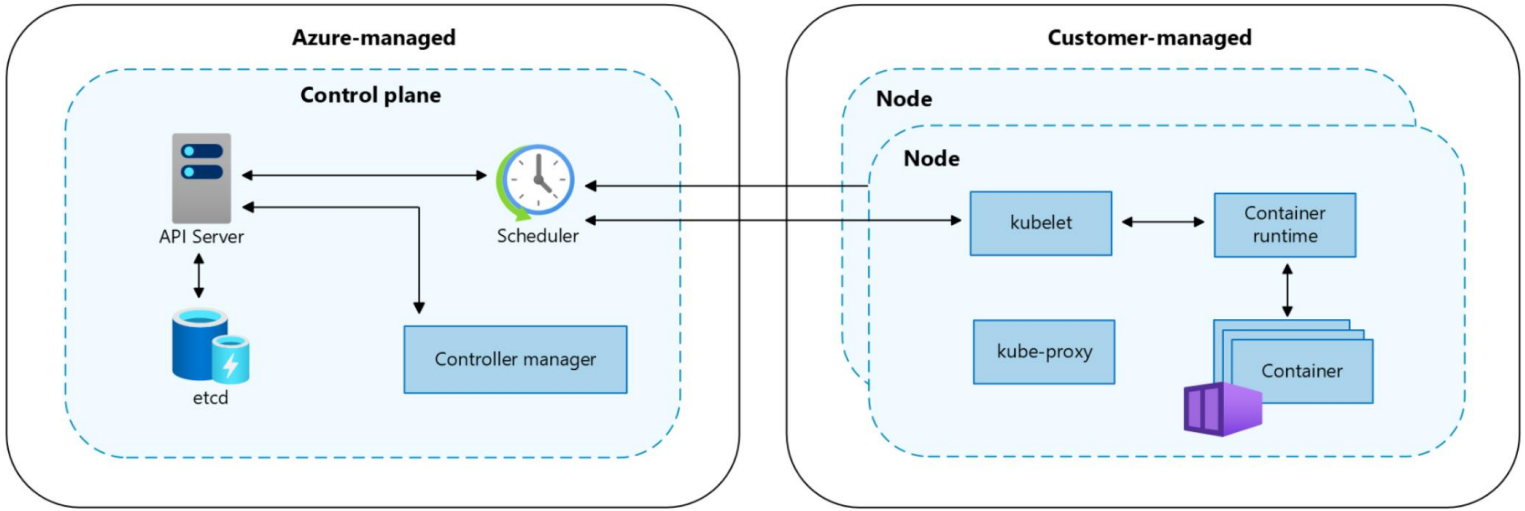
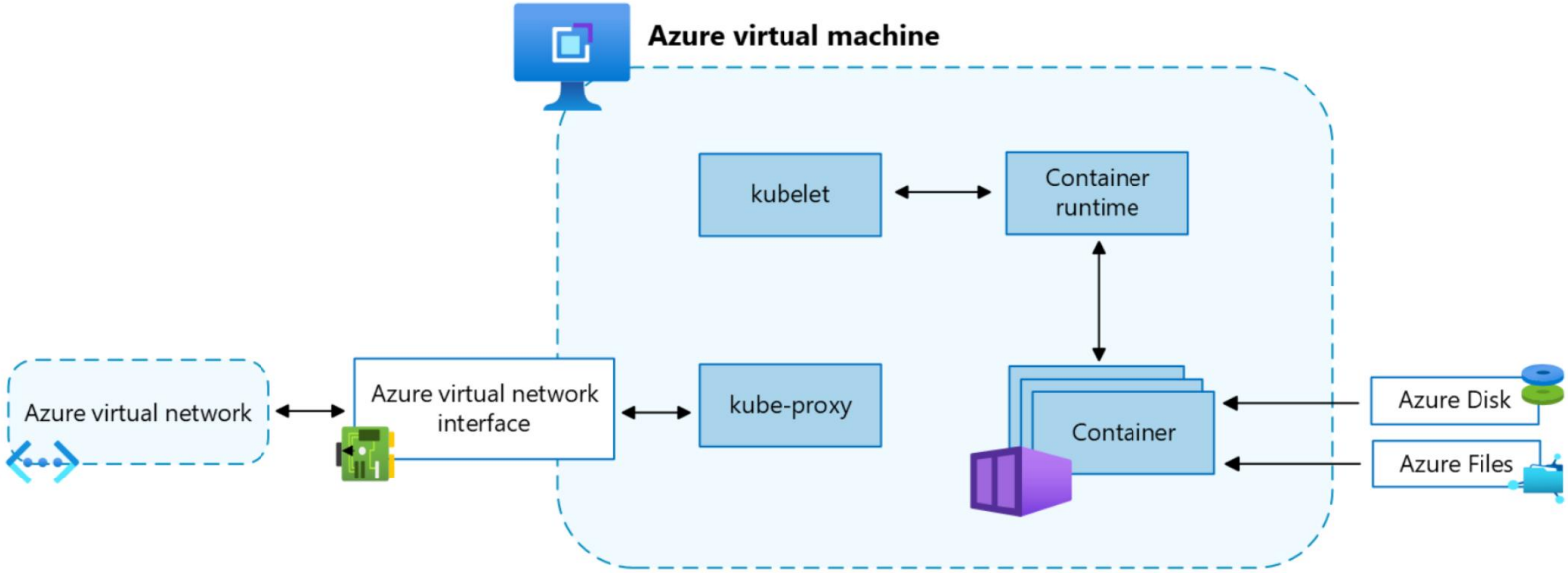
Azure Kubernetes Service (1/2)



Azure Kubernetes Service (2/2)



Azure Kubernetes Service – gestion facile

```
az aks create --resource-group
myResourceGroup --name myAKSCluster
--node-count 3 --node-vm-size
standard_l8s_v3 --generate-ssh-keys
```

az aks upgrade --resource-group `myResourceGroup` --name `myAKSCluster` --kubernetes-version 1.34.4 --control-plane-only

az aks nodepool scale --resource-group <resource-group-name> --cluster-name <cluster-name> --name <node-pool-name> --node-count 5 --no-wait

Plan / Service managés (PaaS)

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- PostgreSQL
- Kubernetes services
- **Machine Learning**

Azure: travaux pratiques

Azure: TP – PostgreSQL "managé" (1/3)

- *Instanciez un **MySQL** Flexible Server*
- *Configurez l'instance (sécurité, accès, performance, backup, redondance, …)*
- *Commencez à utiliser l'instance PostgreSQL (depuis vs-code ?)*

https://github.com/mdimonte/hexagonecloud/blob/main/azure/02-managed-services/README.md

Azure: TP – PostgreSQL "managé" (2/3)

- *Instanciez un cluster **AKS***
- *Démontrez que le cluster Kubernetes est accessible*

https://github.com/mdimonte/hexagonecloud/blob/main/azure/02-managed-services/README.md

Azure: TP – PostgreSQL "managé" (3/3)

- *Instanciez un service **Machine Learning***
- *Réalisez la configuration initiale du service*
- *Explorez comment utiliser ce service*

Découverte de AML

Construire une pipeline AML

https://github.com/mdimonte/hexagonecloud/blob/main/azure/02-managed-services/README.md

Merci :)